Journal of Psychological Inquiry
2023, Vol. 27, No. 2, pp. 49-61
©Great Plains Behavioral Research Association

How message features affect perceptions of the permissibility of medical (and recreational) marijuana

Olivia Tore, Grace Dempsey, Taniya Dsouza, Taylor Ferello, & Tammy L. Sonnentag

Xavier University

Abstract – With the recent shift away from prohibition of marijuana in the United States, the acceptability of marijuana seems to be increasing (Pew Research Center, 2015; Pizzorno, 2016). One possible explanation for more accepting attitudes could be an increase in pro-marijuana messaging and marketing (Berg, 2015). Although previous research has revealed that messages powerfully affect individuals' perceptions of a substance, the types of messages and the extent to which messages affect permissibility of marijuana needs more investigation (see Berg, 2015; Huijding, 2004; Stautz, 2017). Consequently, this study examined if two message features – valence (i.e., pro/positive and anti/negative) and source (i.e., fact-based vs. testimonials from hypothetical marijuana users) – affect the perceived permissibility of *medical* marijuana use. This study also examined if the effects of the message features generalize to perceptions of the permissibility of *recreational* marijuana. Using a 2 x 2 between-subjects experimental design, 116 undergraduate students enrolled in psychology courses at a midsized private university read positively or negatively framed information about the use of medical marijuana from a fact-based, data-driven perspective or from hypothetical marijuana users' perspectives. Results revealed that perceptions of the permissibility of medical marijuana were lower when messages framed information negatively compared to positively – but this effect did not generalize to perceived permissibility of recreational marijuana. No significant effects emerged for source of message. The current study contributes to literature on perceptions of medical (and recreational) marijuana and offers meaningful directions for future research on the topic(s).

In recent years, marijuana has become increasingly available for both medical and recreational uses. Although marijuana is known to be a relatively "safe" drug – with most people who use marijuana *not* experiencing clinical or social problems (Eisen et al., 2002; von Sydow et al., 2001; Wagner & Anthony, 2002) – marijuana consumption still carries some risk of adverse consequences, including accidents and poor psychosocial outcomes (Hall, 2009). For example, there are potentially damaging short- and long-term effects associated with marijuana use. Among younger people (i.e., adolescents), marijuana use is associated with impaired thinking and problem-solving, memory declines, learning difficulties, attention issues, and the risk of addiction (Centers for Disease Control and Prevention, 2021). Similarly, marijuana use among adults has been linked to short-term memory impairments, reduced motor-coordination and reaction time, and challenges making sound judgments (Lenné et al., 2010). Due to the potential negative effects of marijuana use, legalization of the substance (for recreational and medical use) has been extremely controversial. Despite the controversy, United States legislation – and the social attitudes that tend to accompany legislation – have shifted away from prohibition towards legalization in recent years. Specifically, recent research has demonstrated that college students' attitudes towards the permissibility of marijuana usage has increased (Pizzorno, 2016). One explanation for these more permissive attitudes reflects the pro-marijuana marketing that has accompanied the legalization of medical marijuana (in 37 of the 50 states) and recreational marijuana (in 21 of the 50 states; National Conference of State Legislatures [NSCL], 2022). Since marketing can powerfully affect individuals' attitudes toward substances, the current study examined if the

features of messages marketed to individuals affect their attitudes toward marijuana. Specifically, the current study examined if pro/positive or anti/negative messages affect the perceived permissibility of medical and recreational marijuana when the source of the information is based on fact or testimonial.

Power of Messages on Attitudes (and Behaviors)

A robust literature in psychological science documents the power of messages to shape individuals' attitudes (and behaviors). For example, in a recent study by Yeh et al. (2021) examining the social and physical consequences of contracting COVID-19, 440 college students – a demographic group highly resistant to mask wearing – reported their attitude towards wearing a mask after being randomly assigned to read a blog containing messages describing the negative physical consequences of having COVID-19, the negative social consequences of having COVID-19, or the struggles that accompany moving into a new home (control condition). Results revealed that individuals who read blogs with messages describing the physical or social consequences of contracting COVID-19 were more in favor of wearing a mask than individuals who read the control-related blog. These findings demonstrate that messages can impact individuals' health-related attitudes, even individuals who seem relatively resistant to attitude or behavior change. Such results have powerful public health implications.

The public health benefits of using messages to change individuals' attitudes are important, because messages can shape public opinion regardless of whether the individuals actively seek or are incidentally exposed to the content. For example, Lewis and Sznitman (2019) examined whether incidental (i.e., unintentional) exposure to positive information about marijuana affected individuals' attitudes towards marijuana use. In the study, 554 individuals reported their attitude toward the legalization of medical marijuana after reporting whether they actively sought information (i.e., searched for information themselves) or had incidentally encountered information about medical marijuana from various media sources in the previous 30 days. The results revealed that intentionally seeking *and* incidental exposure were both associated with positive attitudes toward medical marijuana, and those positive attitudes were predictive of stronger support for pro-marijuana legislation. Such results suggest that individuals do not need to be actively seeking information about a topic for messages to impact their attitudes. In fact, additional research has demonstrated that exposure to anti-marijuana messages reduces individuals' future intentions to use marijuana among those with positive attitudes toward the message (Alvaro et al., 2013). Such research demonstrates why investigating the effects of messages on attitudes has received significant attention in psychological science.

Factors Affecting the Power of Messages on Attitudes

Although a vast literature documents the various features of messages that are associated with individuals' attitudes, two commonly cited features include the credibility of the source and the positive or negative framing of information in the messages. Among researchers examining the credibility of the source, two powerful sources include scientific/fact-based messages *and* narratives/testimonials.

Arguably, the most obvious credible source of information is science, and research demonstrates that fact-based information meaningfully affects individuals' attitudes. For example, in a study examining the educational value of fact-based messages about nicotine, Parker et al. (2021) randomly assigned 543 individuals to read or not read educational messages about nicotine (e.g., "Nicotine is the addictive substance in tobacco products"). All participants then reported their attitudes toward understanding the harmful effects of nicotine. Results revealed that individuals who read the educational messages, as compared to those who did not, reported more accurate knowledge of nicotine and more favorable attitudes toward understanding the harmful effects of nicotine. Such research is consistent with literature, across a wide range of health-related topics, documenting the power of facts to promote knowledge and affect individuals' attitudes.

Although fact-based information can powerfully affect individuals' attitudes, facts can be perceived as "cold" and "distant", lacking the personal, often emotional, nature of human narratives (De Graaf et al., 2016; Sznitman & Lewis 2015). In fact, researchers argue that narratives (i.e., testimonials) exert strong influence on individuals' attitudes, because narratives offer a personal and emotional perspective (De Graaf et al., 2016; Morris et al., 2019). For example, a recent study examined the effects of patient testimonials about marijuana compared with informationally equivalent non-testimonial (i.e., informational/fact-based) messages. The study revealed that exposure to the testimonials led to significantly more positive attitudes toward marijuana than the informational messages, and these effects were explained by participants feeling more emotionally connected to the patient story (Sznitman & Lewis 2015). Such work is consistent with other research on the effectiveness of narratives (Green & Brock, 2000; Slater & Rouner, 2002; Morris et al., 2019), including

work using patient's successful experiences with medical marijuana as a treatment for a variety of stigmatized (e.g., HIV/AIDS) and non-stigmatized illnesses (e.g., cancer; Lewis & Sznitman, 2017).

As described previously, in addition to the credibility of the source, a second important feature of messages that influences individuals' attitudes is the positive or negative valence of information. A vast multidisciplinary literature has been focused on understanding the purpose of framing which, ultimately, serves to convey content in a positive or negative manner; for example, when content is framed positively, outcomes are gains, whereas when content is framed negatively, outcomes are losses (e.g., Levin, Schneider, & Gaeth, 1998; Tversky & Kahneman, 1981). The negativity principle (see Avdagic & Savage, 2021) suggests that humans have a tendency to attend to more readily (and be influenced by) negative compared to positive information (Baumeister et al. 2001; Tversky & Kahneman 1979; Rozin & Royzman 2001). Examples of the negativity principle are found across many topics, including research demonstrating how public perceptions of the economy are more strongly affected by negative than positive economic news coverage (Soroka 2006); how job placement programs tend to be evaluated more negatively when they are framed in terms of their failure rather than their success rates (Davis & Bobko 1986); and how support for government welfare funding diminishes when information about immigrants is framed negatively compared to positively (Avdagic & Savage, 2021). Additionally, negative information has been demonstrated to play a greater role in attitudes toward U.S. presidential candidates and political parties (Holbrook et al. 2001), with individuals more readily remembering negative political advertising than positive advertising (Johnson-Cartee & Copland 1991).

When the negativity principle has been studied in the context of marijuana, the findings are mixed. Some researchers reveal that negative or anti-marijuana messages increase the perceived harm of the substance (Stevens et al., 2019) and reduce consumption (Alvaro et al., 2013; Palmgreen et al. 2001). However, other researchers have revealed that negative or anti-marijuana messages create a "boomerang effect" and inadvertently improve attitudes and increase individuals' intentions to use marijuana (Czyzewska & Ginsburg, 2007; Yzer et al., 2003). Given the mixed findings in the literature, it is important to continue studying the effects of negative (and positive) messages on perceptions of the permissibility of marijuana. Such research may reveal the conditions under which negative (and positive) messages are most effective in shaping attitudes toward the substance.

Current Study

The purpose of the current study was to examine if individuals' perceptions of the permissibility of medical marijuana is affected by positively or negatively valenced messages from two different sources: scientific fact or personal testimonials (from hypothetical marijuana users). Additionally, the current study examined if the effects of these messages generalize to the perceived permissibility of recreational marijuana. Because marijuana is becoming increasingly common – both medically and recreationally – it is important to understand how messages may affect attitudes toward the substance. The findings could potentially help public health campaigns deliver messages that curb problematic marijuana use. We hypothesized that negatively valenced messages will more adversely impact the perceived permissibility of medical marijuana compared to positive messaging, and this effect will be especially true when the message reflects a personal testimonial (compared to messages with solely science-based fact).

Method

Participants

Participants included 116 undergraduate students (75 women, 37 men, 3 non-binary/non-conforming, 1 no response) ranging in age from 18 to 23 (M = 19.93, SD = 1.17) from Xavier University. The majority of students reported identifying as White (79.3%), with smaller percentages identifying as Black/African-American (9.5%), Biracial (4.3%), Asian (3.4%), Hispanic/Latino (2.5%), and Hawaiian/Pacific Islanders (1%). All participants provided informed consent prior to taking part in the study, which was conducted online using Qualtrics survey software and, therefore, completed the study at a time and location convenient to the participant.

Design

The study was conducted as a 2 (Valence of Message about Marijuana: Pro/Positive vs. Anti/Negative) x 2 (Framing of Information about Marijuana: Fact-Based vs. Hypothetical Marijuana Users) between-subjects factorial design. The dependent variables included the Perceived Permissibility of Medical *and* Recreational Marijuana.

Materials

Marijuana Use Messages

Four vignettes, written for the purposes of the current study (see Appendix), positively or negatively framed information about the use of medical marijuana from a fact-based data-driven perspective or from

hypothetical marijuana users' perspectives. Messages reflecting the fact-based, data-driven perspective were written using current information from credible sources (e.g., Centers for Disease Control and Prevention, 2021; Pew Research Center, 2015) and were accompanied by citations to aid the messages perceived credibility. Messages reflecting the perspective of hypothetical marijuana users describe credible real-world-relevant (positive or negative) experiences with medical marijuana. With the exception of operationalizing the manipulated variables (i.e., valence and pro/positive or anti/negative), the vignettes were written to be as similar as possible (e.g., length, style).

Perceived Permissibility of Medical Marijuana Use

To measure the perceived permissibility of medical marijuana, 23 items from the Clinicians' Attitudes about Opioid Scale were adapted (CAOS; see Wilson et al., 2013). As the title of the CAOS suggests, the original measure asked clinicians to rate their beliefs about the prescription use of *opioid*. For the purposes of the current study, items on the measure were revised to replace *opioid* with *medical marijuana*. Additionally, two items asking clinicians about their specific opioid prescribing behavior were removed. Participants in the current study rated the 23 items on a scale ranging from 1 (*Disagree*) to 5 (*Agree*). After reverse scoring seven negatively keyed items, scores across all items were averaged, with higher scores reflecting greater perceived permissibility of medical marijuana use.

Perceived Permissibility of Recreational Marijuana Use

An adapted version of Simons et al.'s (1998) 25-item Marijuana Motives Measure was used to assess participants' perceptions of the permissibility of recreational marijuana use. The original measure asked individuals about their personal use of recreational marijuana. For the purpose of the current study, the items were revised to ask participants about their perceptions of *others'* use of recreational marijuana. All items were rated on a scale ranging from 1 (*Never*) to 5 (*Always*). Scores across all items were averaged, with higher scores reflecting greater perceived permissibility of recreational marijuana use.

Procedure

Xavier University IRB approval was obtained prior to conducting the present study. The study was conducted online using Qualtrics survey software. After volunteering to participate via the School of Psychology's electronic participant pool system, Sona Systems, participants clicked a secure transfer protocol that directed them to complete an informed consent form in Qualtrics. After providing consent, participants were randomly assigned – using features available in Qualtrics – to one of the study's four marijuana message conditions (i.e., pro- or anti-medical marijuana information from fact-based data-driven perspective or from hypothetical marijuana users' perspectives). After reading the messages, participants completed the Perceived Permissibility of Medical Marijuana and Perceived Permissibility of Recreational Marijuana measures. Subsequently, participants completed a demographic form that asked them about their gender identity, racial/ethnic identity, college year, and age. Finally, participants were debriefed and thanked for their involvement in the study.

Results

To test the hypotheses that negatively-valenced messages would more adversely affect the perceived permissibility of medical and recreational marijuana use than positively-valenced messages, particularly when framed from hypothetical marijuana users' perspectives, separate 2 (Valence of Message about Marijuana: Pro/Positive vs. Anti/Negative) x 2 (Framing of Information about Marijuana: Fact-Based vs. Hypothetical Marijuana Users) between-subjects factorial ANOVAs were conducted.

Perceived Permissibility of Medical Marijuana Use

The main effect of Valence of Message about Marijuana was significant, revealing that participants perceived the use of medical marijuana as less permissible when they read negatively valenced information (M = 2.91, SD = .52) compared to positively valenced information (M = 3.39, SD = .57) about medical marijuana, $F(1, 112) = 23.46$, $p < .001$. The main effect of Framing of Information about Marijuana was not significant, revealing that participants' perceptions of the perceived permissibility of medical marijuana did not differ whether they read information framed from facts (M = 3.24, SD = .62) or marijuana users' perspective (M = 3.07, SD = .56), $F(1, 112) = 2.77$, $p > .05$. The interaction between Valence of Message about Marijuana and Framing of Information about Marijuana was not significant, $F(1, 112) = 2.10$, $p > .05$.

Perceived Permissibility of Recreational Marijuana Use

The main effect of Valence of Message about Marijuana was not significant, revealing that participants' perceptions of the perceived permissibility of recreational marijuana did not differ whether they read positively (M = 2.29, SD = 0.74) compared to negatively (M = 2.24, SD

= 0.74) valenced information about medical marijuana, $F(1, 112) = 0.13$, $p = .72$. The main effect of Framing of Information about Marijuana was also not significant, revealing that participants' perceptions of the perceived permissibility of recreational marijuana did not differ whether they read information framed (about medical marijuana) from facts ($M = 2.28$, $SD = 0.77$) or marijuana users' perspective ($M = 2.26$, $SD = 0.71$), $F(1, 112) = 0.01$, $p = .91$. Finally, the interaction between Valence of Message about Marijuana and Framing of Information about Marijuana was not significant, $F(1, 112) = 0.01$, $p = .95$. Such results suggest that any effects of the messages on perceptions of the permissibility of medical marijuana did *not* extend to perceptions of recreational marijuana.

Discussion

Attempting to add to previous research examining if features of messages affect individuals' perceptions, the current study examined if two message features – valence (i.e., pro/positive and anti/negative) and source (i.e., fact-based vs. testimonials from hypothetical marijuana users) – affect the perceived permissibility of medical marijuana use. The study also examined if the effects of the message features generalize to perceptions of the permissibility of recreational marijuana. Results revealed that perceptions of the permissibility of medical marijuana decreased when messages framed information negatively compared to positively – but this effect did not generalize to perceived permissibility of recreational marijuana.

Individuals who read negatively-valenced messages about medical marijuana reported less permissibility for the substance than individuals who read positively-valenced messages. This finding supports previous research (see Avdagic & Savage, 2021) documenting the powerful role of negatively-valenced information on individuals' attitudes. In the current study, negatively-valenced information about medical marijuana clearly communicated the harmful effects of the substance on individuals' health, and when the negative consequences for a behavior are clear (and there is an easy way to avoid those consequences), research suggests that individuals' avoidance orientations tend to be activated (Lang, 2006). Although individuals' motivations to avoid marijuana were not directly measured in the current study, individuals' approach (appetitive) and avoidance (aversive) orientations provide an excellent direction for future research on the topic. It is possible that such future research may reveal that individuals who experience negative information about medical marijuana as particularly aversive (to themselves and others' health) may perceive the substance as highly impermissible.

Although the current study revealed that negatively-valenced messages about medical marijuana were perceived as less permissible than positively-valenced messages, this effect did not directly generalize to the perceived permissibility of *recreational* marijuana. Prior research suggests that attitudes toward medical marijuana may indirectly, rather than directly, influence attitudes toward recreational marijuana (see Lewis & Sznitman, 2018) via individuals' own attitudes, beliefs, and intentions toward marijuana. Such research suggests that future studies may want to examine more closely the factors that promote the generalization of attitudes about medical marijuana to recreational marijuana. One such factor may be personal experience with marijuana, as non-marijuana users report less accepting attitudes than users (Pearson et al., 2017).

Finally, the current study revealed that two sources of information about medical marijuana (i.e., fact-based vs. testimonials from hypothetical marijuana users) did not yield different levels of perceived permissibility of the substance. Such a result might suggest that both fact-based and testimonial-based information may affect individuals' attitudes toward medical marijuana. Although such a conclusion needs to be supported by additional research (particularly research with a control condition) there is existing literature supporting the effectiveness of both fact-based (see Parker et al., 2021) and narrative-based (i.e., testimonials; Kteily-Hawa et al., 2020; Morris et al., 2019) information on individuals' attitudes. Future research could explore the factors that may make a particular source of information – fact or testimonial – more effective. One such factor may be the number of people (negatively or positively) perceived to be affected by the information. Although testimonials offer a personal and emotional perspective (De Graaf et al., 2016; Morris et al. (2019), they may be perceived as limited in scope – affecting fewer individuals – than fact-based information.

Limitations and Future Directions

Although the current study has many strengths, it is important to note a few limitations. Unfortunately, the sample reflected mostly White, college-aged students from a private university in the Midwest, and these responses are likely not generalizable to individuals from other races/ethnicities, ages, or socioeconomic statuses. The directions for future research mentioned previously would benefit from collecting data from a more diverse sample. Additionally, the current study may have lacked mundane realism, such that reading information about medical marijuana in an online research setting (i.e., Qualtrics) does not closely resemble how individuals

obtain information in the real world. Future research should employ a more realistic methodology – such as exposing people to information via the radio, television, or newspapers – to understand the impact of such information more clearly on their attitudes. For example, Zerhouni et al. (2019) revealed that individuals' passive (i.e., unintended) exposure to information promoted consumption. Such passive exposure is likely more consistent with individuals' real-world experiences and may be a fruitful direction for future research.

References

Alvaro, E. M., Crano, W. D., Siegel, J. T., Hohman, Z., Johnson, I., & Nakawaki, B. (2013). Adolescents' attitudes toward antimarijuana ads, usage intentions, and actual marijuana usage. *Psychology of Addictive Behaviors*, *27*(4), 1027–1035. https://doi.org/10.1037/a0031960.

Avdagic, S., & Savage, L. (2021). Negativity bias: The impact of framing of immigration on welfare state support in Germany, Sweden, and the UK. *British Journal of Political Science, 51*(2), 624–645. https://doi.org/10.1017/S0007123419000395.

Baumeister, R. F., Bratslavsky, E., Finkenauer, C., & Vohs, K. D. (2001). Bad is stronger than good. *Review of General Psychology, 5*(4), 323-370. https://doi.org/10.1037/1089-2680.5.4.323.

Centers for Disease Control and Prevention. (2021, September 8). *Marijuana and public health: Teens*. Centers for Disease Control and Prevention. https://www.cdc.gov/marijuana/health-effects/teens.html.

Czyzewska, M., & Ginsburg, H. J. (2007). Explicit and implicit effects of anti-marijuana and anti-tobacco TV advertisements. *Addictive behaviors*, *32*(1), 114-127. https://doi.org/10.1016/j.addbeh.2006.03.025.

Davis, M. A., & Bobko, P. (1986). Contextual effects on escalation processes in public sector decision making. *Organizational Behavior and Human Decision Processes*, *37*(1), 121-138. https://doi.org/10.1016/0749-5978(86)90048-8

De Graaf, A., Sanders, J., & Hoeken, H. (2016). Characteristics of narrative interventions and health effects: A review of the content, form, and context of narratives in health-related narrative persuasion research. *Review of Communication Research*, *4*(1), 88-131. https://doi.org/10.12840/issn.2255-4165.2016.04.01.011.

Eisen, S. A., Chantarujikapong, S., Xian, H., Lyons, M. J., Toomey, R., True, W. R., Scherrer, J. F., & Tsuang, M. T. (2002). Does marijuana use have residual adverse effects on self-reported health measures, socio-demographics, and quality of life? A monozygotic co-twin control study in men. *Addiction*, *97*(9), 1137-1144. https://doi.org/10.1046/j.1360-0443.2002.00120.x.

Green, M. C. & Brock, T. C. (2000). The role of transportation in the persuasiveness of public narratives. *Journal of Personality and Social Psychology*, *79*(5), 701-725. https://doi.org/10.1037/0022-3514.79.5.701.

Hall, W. (2009). The adverse health effects of cannabis use: What are they, and what are their implications for policy. *International Journal of Drug Policy*, *20*(6), 458-466. https://doi.org/10.1016/j.drugpo.2009.02.013.

Holbrook, A. L., Krosnick, J. A., Visser, P. S., Gardner, W. L., & Cacioppo, J. T. (2001). Attitudes toward presidential candidates and political parties: Initial optimism, inertial first impressions, and a focus on flaws. *American Journal of Political Science 45*(4), 930-950. https://doi.org/10.2307/2669333.

Huijding, J., de Jong, P. J., Wiers, R. W., & Verkooijen, K. (2004). Implicit and explicit attitudes toward smoking in a smoking and a nonsmoking setting. *Addictive Behaviors, 30*(5), 949-961. https://doi.org/10.1016/j.addbeh.2004.09.014.

Johnson-Cartee, K. S. & Copeland, G. A. (1991). *Negative political advertising: Coming of age*. Lawrence Erlbaum Associates, Inc.

Kteily-Hawa, R., Hari, S., Soor, J. K., Wong, J. P.-H., Chikermane, V., Chambers, L. A., & Vahabi, M. (2020). Paradigm shifts in sexual health: Quantitative analysis of story and fact-based health education interventions. *Canadian Journal of Human Sexuality*, *29*(1), 45–56. https://doi.org/10.3138/cjhs.2018-0037.

Lang, A. (2006). Using the limited capacity model of motivated mediated message processing to design effective cancer communication messages. *Journal of Communication, 56*(1), 57-80. https://doi.org/10.1111/j.1460-2466.2006.00283.x.

Lenné M. G., Dietze, P. M., Triggs, T.J., Walmsley, S., Murphy B., & Redman, J.R. (2010). The effects of cannabis and alcohol on simulated arterial driving: Influences of driving experience and task demand. *Accident Analysis & Prevention, 42*(3), 859-866. https://doi.org/10.1016/j.aap.2009.04.021.

Levin, I. P., Schneider, S. L., & Gaeth, G. J. (1998). All frames are not created equal: A typology and critical analysis of framing effects. *Organizational Behavior and Human Decision Processes, 76*(2), 149–188. https://doi.org/10.1006/obhd.1998.2804.

Lewis, N., & Sznitman, S. R. (2017). You brought it on yourself: The joint effects of message type, stigma, and responsibility attribution on attitudes toward medical cannabis. *Journal of Communication*, *67*(2), 181-202. https://doi.org/10.1111/jcom.12287.

Lewis, N., & Sznitman, S. R. (2018). Examining effects of medical cannabis narratives on beliefs, attitudes, and intentions related to recreational cannabis: A web-based randomized experiment. *Drug and Alcohol Dependence, 185*(1), 219–225. https://doi.org/10.1016/j.drugalcdep.2017.11.028.

Lewis, N., & Sznitman, S. R. (2019). Engagement with medical cannabis information from online and mass media sources: Is it related to medical cannabis attitudes and support for legalization. *International Journal of Drug Policy, 73*(1), 219–227. https://doi.org/10.1016/j.drugpo.2019.01.005.

Morris, B. S., Chrysochou, P., Christensen, J. D., Orquin, J. L., Barraza, J., Zak, P. J., & Mitkidis, P. (2019). Stories vs. facts: Triggering emotion and action-taking on climate change. *Climatic Change*, *154*(1), 19–36. https://doi.org/10.1007/s10584-019-02425-6.

National Conference of State Legislatures. (2022, November 9). *State medical cannabis laws*. https://www.ncsl.org/research/health/state-medical-marijuana-laws.aspx.

Palmgreen, P., Donohew, L., Lorch, E. P., Hoyle, R. H., & Stephenson, M. T. (2001). Television campaigns and adolescent marijuana use: Tests of sensation seeking targeting. *American Journal of Public Health*, *91*(2), 292-302. https://doi.org/10.2105/AJPH.91.2.292.

Parker, M. A., Byers, J. E., & Villanti, A. C. (2021). Effect of brief nicotine corrective messaging on nicotine beliefs in persons who use opioids. *Experimental and Clinical Psychopharmacology, 29*(2), 1-7. https://doi.org/10.1037/pha0000497.

Pearson, M. R., Liese, B. S., Dvorak, R. D., & Marijuana Outcomes Study Team. (2017). College student marijuana involvement: Perceptions, use, and consequences across 11 college campuses. *Addictive Behaviors*, *66*(1), 83-89. https://doi.org/10.1016/j.addbeh.2016.10.019.

Pew Research Center (2015). Two-thirds of Americans support marijuana legalization. https://www.pewresearch.org/fact-tank/2019/11/14/americans-support-marijuana-legalization/

Pizzorno, J. (2016). What should we tell our patients about marijuana (Cannabis indica and Cannabis sativa). *Integrative Medicine 15*(6), 8–12.

Rozin, P. & Royzman, E. B. (2001). Negativity bias, negativity dominance, and contagion. *Personality and Social Psychology Review, 5*(4), 296–320. https://doi.org/10.1207/S15327957PSPR0504_2.

Simons, J., Correia, C. J., Carey, K. B., & Borsari, B. E. (1998). Validating a five-factor marijuana motives measure: Relations with use, problems, and alcohol motives. *Journal of Counseling Psychology, 45*(3), 265-273. https://doi.org/10.1037/0022-0167.45.3.265.

Slater, M. D & Rouner, D. (2002). Entertainment-education and elaboration likelihood: Understanding the processing of narrative persuasion. *Communication Theory, 12*(2), 173-191. https://doi.org/10.1111/j.1468-2885.2002.tb00265.x.

Soroka, S. N. (2006). Good news and bad news: Asymmetric responses to economic information. *Journal of Politics, 68*(2), 372-385. https://doi.org/10.1111/j.1468-2508.2006.00413.x.

Stautz, K., Frings, D., Albery, I. P., Moss, A. C., & Marteau, T. M. (2017). Impact of alcohol-promoting and alcohol-warning advertisements on alcohol consumption, affect, and implicit cognition in heavy-drinking young adults: A laboratory-based randomized controlled trial. *British Journal of Health Psychology*, *22*(1), 128–150. https://doi.org/10.1111/bjhp.12221.

Stevens, E. M., Cohn, A. M., Villanti, A. C., Leshner, G., Wedel, A., & Wagener, T. L. (2019). Perceived effectiveness of anti-marijuana messages in adult users and nonusers: An examination of responses to messages about marijuana's effects on cognitive performance, driving, and health. *Journal of Studies on Alcohol and Drugs*, *80*(4), 415-422. https://doi.org/10.15288/jsad.2019.80.415.

Sznitman, S. R., & Lewis, N. (2015). Is cannabis an illicit drug or a medicine? A quantitative framing analysis of Israeli newspaper coverage. *International Journal of Drug Policy*, *26*(5), 446-452. https://doi.org/10.1016/j.drugpo.2015.01.010.

Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science*, *211*(4481), 453–458. https://doi.org/10.1126/science.7455683.

von Sydow, K., Lieb, R., Pfister, H., Höfler, M., Sonntag, H., & Wittchen, H. U. (2001). The natural course of cannabis use, abuse, and dependence over four years: a longitudinal community study of adolescents and young adults. *Drug and Alcohol Dependence*, *64*(3), 347-361. https://doi.org/10.1016/S0376-8716(01)00137-5.

Wagner, F. A., & Anthony, J. C. (2002). From first drug use to drug dependence: developmental periods of risk for dependence upon marijuana, cocaine, and alcohol. *Neuropsychopharmacology*, *26*(4), 479-488. https://doi.org/10.1016/S0893-133X(01)00367-0.

Wilson, H.D., Dansie, E.J., Kim, M.S., Moskvitz, B.L., Chow, W., & Turk, D.C. (2013). Clinicians' attitudes and beliefs about opioids survey (CAOS): Instrument development and results of a national physician survey. *The Journal of Pain, 14*(6), 613-627. https://doi.org/10.1016/j.jpain.2013.01.769.

Yeh, M. A., Mirabito, A. M., & Finkelstein, S. R. (2021). Physical risk messaging enhances favorable attitudes toward mask wearing. *Journal of Consumer Affairs*, *55*(4), 1682–1690. https://doi.org/10.1111/joca.12402.

Yzer, M. C., Cappella, J. N., Fishbein, M., Hornik, R., Sayeed, S., & Ahern, R. K. (2004). The role of distal variables in behavior change: effects of adolescents' risk for marijuana use on intention to use marijuana *Journal of Applied Social Psychology*, *34*(6), 1229-1250. https://doi.org/10.1111/j.1559-1816.2004.tb02005.x.

Zerhouni, O., Bègue, L., & O'Brien, K. S. (2019). How alcohol advertising and sponsorship works: Effects through indirect measures. *Drug and Alcohol Review, 38*(4), 391–398. https://doi.org/10.1111/dar.12929.

Author Note
We would like to thank Dr. Kyle Stephenson for his feedback on the initial ideas for this study.

Appendix – Positive Fact-Based

Please read through the information below at least twice and use the information provided to evaluate your own perspectives. You will not be able to move to the next page until at least ***three minutes have elapsed****.*

Since 1996, medical marijuana (or cannabis) has been legalized in 36 U.S. states and has been recognized as a legitimate treatment for various medical conditions by credible healthcare professionals (Timmerman 2011). One of the most notable uses of medical cannabis is to manage pain, especially for individuals who suffer from chronic pain due to conditions such as endometriosis (which causes extreme cramps in the pelvic region), multiple sclerosis (which causes severe nerve pain), and fibromyalgia (which causes widespread muscular aches) (Peters 2020). Marijuana can also be used to treat or reduce the symptoms related to Parkinson's Disease, Hepatitis C, epilepsy, diabetes, Alzheimer's Disease, as well as general afflictions of nausea, appetite loss, and muscle spasms (Peters 2020). Additionally, physicians have prescribed marijuana as an effective treatment for alleviating symptoms caused by some mental illnesses, such as reducing obsessive urges or intrusive thoughts for those with Obsessive-Compulsive Disorder, minimizing hypervigilance among those with Posttraumatic Stress Disorder, and managing symptoms associated with autism, depression, anxiety, Bipolar Disorder, and Attention Deficit Hyperactivity Disorder (Chang 2019). Even among people who are not prescribed marijuana as their *main* form of treatment, it has proven to be a valuable substance for lessening side effects from other medications or treatments, such as reducing nausea from chemotherapy in cancer patients (Garcia 2019). Compared to opioids and other types of drug treatments, marijuana is a much safer option because overdosing is *not* possible and it does *not* contain an addictive element, so it is more practical for long-term use (Ochsner 2013). Overall, marijuana has proven to be a very versatile and medically beneficial substance.

Appendix Continued – Negative Fact-Based

Please read through the information below at least twice and use the information provided to evaluate your own perspectives. You will not be able to move to the next page until at least ***three minutes have elapsed****.*

Since 1996, medical marijuana (or cannabis) has been legalized in 36 U.S. states, yet it has been recognized as objectively harmful to the health of its users as well as to the people around them. Scientific evidence has demonstrated that smoking one marijuana joint is as damaging to the lungs as five tobacco cigarettes and can contain up to five times the amount of carbon monoxide and three times the amount of tar that is contained in tobacco, regardless of whether the marijuana is prescribed by a physician or not (Timmerman 2011). Smoking marijuana has also been demonstrated to cause damage to lung tissue and permanent respiratory issues, effects that are also observed through secondhand smoke. Consuming cannabis in any form can contribute to lower bone density, an increased risk of heart disease or heart attacks, and birth defects in fetuses (Chang 2019). Regular long-term marijuana use can similarly lead to illnesses, such as Cannabinoid Hyperemesis Syndrome, which is a condition that causes users to experience regular cycles of severe nausea, vomiting, and dehydration, and can require emergency medical attention and hospitalization (Garcia 2019). Using marijuana can also result in detrimental cognitive effects, including problems with memory, coordination and movement, the ability to think clearly, and the ability to pay attention (Peters 2020). For individuals who are under the age of 25, whose brains are still developing, these side effects can be heightened and lead to irreversible harm to the body (Peters 2020). Marijuana users can also experience intense feelings of anxiety, paranoia, or even thoughts of suicide, and this is more likely among those who have been previously diagnosed with a mental disorder or regularly experience these states (Ochsner 2013). With the increased prevalence of medical marijuana, children, pets, and even some adults have mistakenly consumed marijuana resulting in more frequent trips to the emergency room (Sandre 2021). Overall, marijuana has proven to be a generally harmful substance.

Appendix Continued – Positive Testimonial

Please read through the information below at least twice and use the information provided to evaluate your own perspectives. You will not be able to move to the next page until at least ***three minutes have elapsed****.*

"My uncle smokes weed pretty regularly for his PTSD. He served in Afghanistan and ever since he got back, he's had nightmares and flashbacks that caused panic attacks. He couldn't keep a job and rarely went out because random loud noises triggered his flashbacks. He goes to therapy for it, and he had been taking medicine for a while which helped a little bit, but once he got his medical marijuana card, he got a lot better. The loud noises don't affect him much anymore, and he now only has nightmares occasionally."

- Julie, 22

"I have been on and off different medications for my OCD since I was in middle school, and nothing has worked as well as my prescription marijuana patches. I just put one of them on my arm every day and it seems to be the only way to reduce my urges. One of my main obsessions is thinking that all of my food has expired or has gone bad. It was so bad at one point, that I used to go to the grocery store multiple times a week to replace day-old food in my pantry. It was costing me and my family thousands of dollars and was honestly ruining my life. Since I have started using the patches, I feel like I have more control over my own body, and I only go to the grocery once a week now."

- Mitch, 19

"I have multiple sclerosis and I have a medical marijuana card for the pain. I get these horrible migraines and feel extremely dizzy, and I suffer from other neurological issues as well. Some days, I was so nauseous that I couldn't even eat, so my doctor recommended that I apply for a card because other treatments just weren't working. At first, I was hesitant, but it was so hard being without an appetite for days at a time, so I applied, and I was officially registered within a few weeks. Using cannabis in my treatment has helped a lot with the migraines, and it even seemed to calm down some of my more serious symptoms like trouble speaking and muscle spasms. It isn't a cure-all, but I don't think I could manage the pain without it."

- Suzzanne, 56

Appendix Continued – Negative Testimonial

Please read through the information below at least twice and use the information provided to evaluate your own perspectives. You will not be able to move to the next page until at least ***three minutes have elapsed****.*

"I have been taking standard anxiety medicine for about 5 years, but I was still experiencing a lot of symptoms, so I reached out to my doctor to talk about a medical marijuana card. She agreed that it may help, and I applied for a card and began taking edibles to soothe my anxiety. For the few times I took them, I really just felt more anxious and paranoid than I already was. I remember one-time last year after I ate one, I felt so confused and like I was dying or something, and I just ended up throwing up. I usually just zoned out for a couple of hours because I felt so lazy and unmotivated, and I didn't think it helped with my anxiety either. I stopped taking them a few months ago because I would rather just take my normal anxiety medicine by itself and not have to worry about all of the issues that the edibles caused."

- Sawyer, 20

"I am usually hesitant when prescribing patients medical marijuana. Over the past few years with the legalization of it in different states, it seems to have become more and more acceptable to prescribe a form of cannabis as an initial treatment rather than as a last resort, rather than waiting to understand if other treatment plans are unsuccessful. For many common diseases and disorders, there are clinically tested, efficacious, treatments that have been backed by decades of research, which is not true for medical cannabis, and that is why I am so hesitant to jump on the medical marijuana bandwagon. Not to mention, dispensaries can be careless with their dosages, which is completely unacceptable for any other drug, so it is extremely frustrating as a healthcare professional to see that marijuana isn't held to the same standards as other medicines. Over the past few years, I have seen a lot of adverse reactions within patients that use medical marijuana, often it worsens their symptoms. In patients who have not struggled with anxiety before, I see them coming in with extreme anxiety and paranoia wondering what has happened to them. People that smoke medical marijuana often have issues with asthma and various other lung conditions that can range from mild to severe. With all of these unknown issues and a lack of longitudinal research, I avoid prescribing my patients marijuana.

- Nick, 39

"One of my close friends who had a medical marijuana card frequently hit a dab pen to manage her chronic pain from her arthritis. About a year and a half ago, she was diagnosed with CHS (Cannabinoid Hyperemesis Syndrome) after she went to the ER because she couldn't stop throwing up. Her parents told me that she could have died from dehydration and kidney failure. Even after she quit, she would get these horrible stomach aches and would vomit so

frequently that her throat would burn from too much stomach acid. She doesn't have much of an appetite these days and has lost a lot of weight from her condition. I began to look into CHS because I had never heard of it before, and I found so many stories from others who have had even worse experiences. Now I am terrified of smoking weed after seeing and hearing what she and many others have gone through."

- CLAIRE, 25